Comparative Performance Analysis of Pre-trained Models for Deepfake Audio Detection using Ensemble Learning

Research Paper — Audio Deepfake Detection

Dataset: garystafford/deepfake-audio-detection (HuggingFace)

Framework: PyTorch · HuggingFace Transformers · Gradio

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Abstract

The proliferation of synthetic voice technologies poses a growing threat to digital media authenticity and personal security. This paper presents a comparative performance analysis of ten pre-trained transformer-based audio models evaluated on a binary deepfake audio detection task. Using the publicly available HuggingFace dataset garystafford/deepfake-audio-detection — comprising 1,866 real and synthetic audio samples — we sampled a balanced subset of 100 clips (50 real, 50 fake) sampled at 16 kHz and clipped to five seconds. Each model was loaded via the HuggingFace Transformers pipeline API and assessed using a novel score-based threshold tuning strategy that addresses the fundamental mismatch between general-purpose audio model output spaces and the binary real/fake classification requirement. Two scoring strategies — confidence-based and entropy-based thresholding — were evaluated and the better-performing approach was retained per model. A weighted ensemble was then constructed using F1-score-proportional voting across all ten models. Evaluation metrics included accuracy, precision, recall (sensitivity), specificity, and F1 score. The Audio Spectrogram Transformer fine-tuned on AudioSet (MIT/ast-finetuned-audioset-10-10-0.4593) emerged as the best individual model with an accuracy of 87% and an F1 score of 0.8762, outperforming all other architectures by a substantial margin. The ensemble, while achieving perfect recall (1.00), underperformed the best individual model in overall accuracy (55%), highlighting the challenges of combining models that were not trained for the target task. The system was deployed as a live inference application using Gradio. This work demonstrates the feasibility and limitations of repurposing pre-trained audio models for deepfake detection without fine-tuning.

Keywords: *deepfake audio detection, transformer models, ensemble learning, threshold tuning, wav2vec2, HuBERT, WavLM, Audio Spectrogram Transformer, binary classification*

1. Introduction

1.1 What Are Deepfakes?

Deepfakes refer to artificially generated or manipulated media — including images, video, and audio — produced using advanced machine learning techniques, most notably generative adversarial networks (GANs) and neural text-to-speech (TTS) systems. The term was first coined in 2017 and has since expanded to encompass a wide range of synthetic media generation techniques. Audio deepfakes, in particular, involve the synthesis or cloning of human voices to produce speech that sounds indistinguishable from a genuine speaker, often without that speaker's knowledge or consent.

Modern voice synthesis technologies such as WaveNet, Tacotron, FastSpeech, and VALL-E are capable of generating high-fidelity speech from minimal audio samples. When weaponized maliciously, these technologies enable fraud, impersonation, disinformation campaigns, and identity theft at an unprecedented scale. The growing accessibility of these tools — many available as open-source software or commercial APIs — means that the barrier to producing convincing audio deepfakes has never been lower.

1.2 Why Audio Deepfake Detection Matters

The social and economic consequences of audio deepfakes are significant. In documented cases, synthetic voice recordings have been used to authorize fraudulent financial transactions, generate fake political statements, and create non-consensual intimate audio content. As synthetic audio quality continues to improve, perceptual detection by humans becomes increasingly unreliable, necessitating automated, machine-learning-driven detection pipelines.

Automated deepfake audio detection systems must be accurate, fast, and generalizable across speakers, languages, and synthesis techniques. Current state-of-the-art approaches leverage self-supervised speech representations — particularly transformer-based models pre-trained on large speech corpora — as feature extractors. However, most such models are designed for tasks such as emotion recognition, keyword spotting, or speech classification, not explicitly for deepfake detection. This raises the critical research question: can pre-trained audio models, repurposed without fine-tuning, detect deepfake audio through intelligent score-based inference strategies?

This paper addresses this question through a systematic empirical evaluation of ten pre-trained transformer audio models on a balanced deepfake audio detection benchmark. Our contributions include: (1) a robust score-based threshold tuning methodology that circumvents label space mismatch; (2) a comparative analysis of ten diverse model architectures; (3) a weighted ensemble framework; and (4) a live Gradio deployment of the best model.

2. Background

2.1 Pre-trained Audio Transformer Models

The self-supervised learning paradigm has transformed audio representation learning. Models such as wav2vec 2.0 (Baevski et al., 2020), HuBERT (Hsu et al., 2021), and WavLM (Chen et al., 2022) are trained on hundreds of thousands of hours of unlabeled speech through contrastive or masked prediction objectives, learning rich latent representations that transfer effectively to downstream tasks. These models adopt the transformer architecture originally proposed by Vaswani et al. (2017) and operate directly on raw audio waveforms sampled at 16 kHz, eliminating the need for hand-crafted acoustic features such as Mel-frequency cepstral coefficients (MFCCs).

The Audio Spectrogram Transformer (AST; Gong et al., 2021) takes a different approach, treating audio as two-dimensional time-frequency spectrograms and processing them using a Vision Transformer (ViT) architecture originally developed for image classification. AST fine-tuned on AudioSet-2M — a large-scale audio event detection dataset — captures broad environmental and acoustic patterns that may prove useful for distinguishing synthetic from natural audio.

2.2 The HuggingFace Ecosystem

HuggingFace (Hugging Face Inc., 2023) provides a unified platform for sharing, loading, and deploying machine learning models and datasets. The Transformers library offers a high-level pipeline API that enables the loading of any supported model with minimal code, abstracting away preprocessing, tokenization, and inference. The datasets library provides efficient streaming access to thousands of publicly hosted datasets. Together, these tools enable rapid prototyping and benchmarking of multiple models across standardized evaluation protocols — a capability central to this study.

2.3 Audio Classification Fundamentals

Audio classification involves assigning a categorical label to an audio segment based on its acoustic content. In the context of deepfake detection, this is a binary task: the model must output either real (label 0) or fake (label 1). Standard pipeline approaches involve (1) loading raw audio at a fixed sampling rate, (2) passing it through a feature extractor (e.g., a feature normalizer or spectrogram converter), (3) forwarding through a transformer encoder, and (4) applying a linear classification head over the pooled representation. Performance is assessed using standard binary classification metrics including accuracy, precision, recall, specificity, and F1 score.

3. Methodology

3.1 Dataset Description

The dataset used in this study is garystafford/deepfake-audio-detection, hosted on HuggingFace. The full dataset comprises 1,866 audio samples organized in a binary format — 933 real and 933 fake recordings — with each sample labeled under a single label column (0 = real, 1 = fake). Audio files are loaded by the HuggingFace datasets library and resampled to 16,000 Hz using the Audio feature with the cast_column method.

For computational efficiency — particularly relevant given GPU memory constraints and the sequential loading of ten large models — a balanced subset of 100 samples was selected: 50 real

and 50 fake. Samples were randomly selected using a fixed seed (numpy seed = 42) for reproducibility. Each audio array was clipped to a maximum length of 5 seconds (80,000 samples at 16 kHz) and short clips below 1,600 samples were zero-padded. This preprocessing ensures consistent input dimensions across all models.

3.2 Preprocessing Pipeline

The preprocessing pipeline follows these steps: (1) Loading the dataset using load_dataset with split='train'; (2) Casting the audio column to 16 kHz using Audio(sampling_rate=16000); (3) Stratified sampling — 50 samples per class selected randomly; (4) Array extraction: each sample's audio array is converted to a float32 NumPy array; (5) Length normalization: arrays are clipped to 80,000 samples (5 seconds) or zero-padded if shorter than 1,600 samples. No spectral feature extraction (e.g., MFCC, Mel spectrogram) is performed manually, as each model's feature extractor handles this internally during inference.

3.3 Model Selection

Ten pre-trained models spanning diverse architectures and pre-training tasks were selected for evaluation:

- M1 — facebook/wav2vec2-base: A foundational self-supervised speech representation model trained via contrastive learning on LibriSpeech 960h.
- M2 — superb/wav2vec2-base-superb-er: Wav2Vec2-base fine-tuned for speech emotion recognition on the IEMOCAP dataset.
- M3 — MIT/ast-finetuned-audioset-10-10-0.4593: Audio Spectrogram Transformer fine-tuned on AudioSet for 527-class audio event detection.
- M4 — facebook/hubert-base-ls960: HuBERT trained on LibriSpeech with masked cluster prediction, a powerful speech representation alternative to Wav2Vec2.
- M5 — ntu-spml/distilhubert: A distilled, lightweight version of HuBERT with approximately 75% fewer parameters.
- M6 — microsoft/wavlm-base: WavLM trained with denoising pre-training, adding robustness to noisy and occluded speech conditions.
- M7 — superb/hubert-base-superb-er: HuBERT fine-tuned for emotion recognition under the SUPERB benchmark.
- M8 — ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition: XLS-R large (1.27 GB), a cross-lingual Wav2Vec2 fine-tuned on English emotion data.
- M9 — superb/wav2vec2-base-superb-ic: Wav2Vec2-base fine-tuned for intent classification under SUPERB.
- M10 — superb/wav2vec2-base-superb-ks: Wav2Vec2-base fine-tuned for keyword spotting under the SUPERB benchmark.

3.4 Prediction Pipeline and Score Extraction

Each model was loaded using HuggingFace's pipeline API with task='audio-classification', return_all_scores=True, and truncation=True. For each audio sample, the pipeline returns a list of label-score pairs covering the full output label space of the model. Since none of these models were

trained for a binary real/fake task, their label spaces are domain-specific (e.g., emotion categories, AudioSet event labels, keyword classes) and cannot be directly mapped to the deepfake detection target.

To address this fundamental mismatch, two proxy score signals were computed per sample: (1) Confidence score — the maximum probability across all output classes, which serves as a measure of how definitively the model classifies the sample. (2) Entropy score — the Shannon entropy of the full probability distribution, which quantifies prediction uncertainty. High entropy indicates that the model is uncertain, potentially signaling unusual or synthetic audio characteristics.

3.5 Threshold Tuning

Given that raw scores cannot be interpreted as real/fake probabilities without task-specific training, a threshold tuning procedure was designed to find the optimal binary decision boundary. The tuning explores three strategies: (1) Percentile-based thresholds: thresholds are set at the 10th through 90th percentiles of observed scores in 5% increments, both in original and inverted order; (2) Fixed thresholds: values from 0.1 to 0.9 in 0.05 increments are evaluated; (3) Median split: the sample median is used as the threshold.

For each candidate threshold, binary predictions are generated and the F1 score (for the fake class) is computed. The threshold yielding the highest F1 score while ensuring both classes are predicted is retained. Both confidence-based and entropy-based scoring strategies are evaluated independently, and the strategy (confidence or entropy) that yields the higher F1 is selected per model. This adaptive approach ensures that single-class prediction failures — where all samples are assigned to one class — are detected and mitigated. For the best model (M3/AST), the optimal threshold was identified at 0.80.

3.6 Weighted Ensemble

After individual model evaluation, a weighted ensemble was constructed using F1-score-proportional voting. Each model's prediction weight was computed as its F1 score divided by the sum of all F1 scores. A minimum floor weight of 0.01 was applied to prevent any model from receiving zero weight (which would occur if a model achieved F1 = 0). For each audio sample, the weighted sum of binary predictions was computed across all ten models. A sample was classified as fake if the weighted vote exceeded 0.5. This ensemble aims to leverage the complementary decision boundaries of diverse models while down-weighting poorly performing ones.

4. Evaluation Metrics

The following metrics were used to evaluate both individual models and the ensemble. All metrics are computed with respect to the fake class (positive class = 1).

Accuracy

Accuracy measures the proportion of correctly classified samples out of the total. It is defined as (TP + TN) / (TP + TN + FP + FN). While intuitive, accuracy can be misleading in imbalanced datasets; in this study the balanced 50/50 split makes it a reliable primary metric.

Precision

Precision (also called Positive Predictive Value) measures what fraction of samples predicted as fake are actually fake: TP / (TP + FP). High precision indicates a low false alarm rate.

Recall (Sensitivity)

Recall measures what fraction of actual fake samples are correctly detected: TP / (TP + FN). In security-critical deepfake detection, recall is particularly important — a missed fake (false negative) can have serious consequences.

Specificity

Specificity measures how well the model correctly identifies real audio: TN / (TN + FP). Low specificity means many real samples are falsely flagged as fake (false positive rate is high).

F1 Score

The F1 score is the harmonic mean of precision and recall: 2 * (Precision * Recall) / (Precision + Recall). It provides a single balanced measure that penalizes extreme imbalances between precision and recall, making it the primary optimization target in this study.

5. Results and Discussion

5.1 Performance Comparison Table

Table 1 presents the complete evaluation results for all ten individual models and the ensemble. Values reported are computed from 100 balanced test samples (50 real, 50 fake).

Table 1: Comparative Performance of All Models and Ensemble

Model / System	Architecture	Accuracy	Precision	Recall	Specificity	F1 Score
M1 – wav2vec2-base	Wav2Vec2	0.60	0.5556	1.00	0.20	0.7143
M2 – wav2vec2-superb-er	Wav2Vec2	0.51	0.5051	1.00	0.02	0.6711
M3 – AST-AudioSet	AST Transformer	0.87	0.8364	0.92	0.82	0.8762
M4 – HuBERT-base-ls960	HuBERT	0.59	0.5529	0.94	0.24	0.6963
M5 – DistilHuBERT	HuBERT (distilled)	0.54	0.5222	0.94	0.14	0.6714
M6 – WavLM-base	WavLM	0.55	0.5294	0.90	0.20	0.6667

Model / System	Architecture	Accuracy	Precision	Recall	Specificity	F1 Score
M7 – HuBERT-superb-er	HuBERT	0.52	0.5102	1.00	0.04	0.6757
M8 – wav2vec2-xlsr-emotion	Wav2Vec2 XLS-R	0.63	0.5867	0.88	0.38	0.7040
M9 – wav2vec2-superb-ic	Wav2Vec2	0.64	0.6000	0.84	0.44	0.7000
M10 – wav2vec2-superb-ks	Wav2Vec2	0.62	0.5750	0.92	0.32	0.7077
Weighted Ensemble	Ensemble (all 10)	0.55	0.5263	1.00	0.10	0.6897

** Best individual result per metric highlighted. Ensemble shown for comparison.*

5.2 Key Insights

- M3 (MIT/AST fine-tuned on AudioSet) was the best-performing individual model by a wide margin — accuracy 87%, F1 0.8762 — using an entropy-based strategy at an inverted 55th percentile threshold. Its superior performance stems from its spectrogram-based architecture that captures frequency-domain patterns distinguishing natural speech from synthesized audio.
- M2 (wav2vec2-superb-er) was the worst-performing individual model, with accuracy of 51% (near-random) and specificity of just 0.02. It correctly classified nearly all samples as fake, but failed completely to identify real audio, indicating its emotion recognition features provide almost no discriminative signal for this task.
- Threshold tuning was essential across all models. Without adaptive thresholding, the majority of models defaulted to predicting a single class (typically all-fake), resulting in degenerate metrics. The percentile-based strategy successfully recovered meaningful predictions in all ten cases.
- The weighted ensemble achieved perfect recall (1.00) but poor specificity (0.10) and accuracy (55%). This reflects a conservative ensemble bias toward classifying samples as fake — driven by the dominance of models with high recall but low specificity in the weight distribution.
- M3 outperformed the ensemble on every metric except recall. This counterintuitive result (where the best individual beats the combination) occurs because ensemble diversity is beneficial only when component models contribute complementary, accurate information — here, many models are near-random, diluting M3's strong signal.
- Larger models did not necessarily outperform smaller ones. M8 (XLS-R, 1.27 GB) achieved F1 of 0.7040, performing similarly to M1 (Wav2Vec2-base, 380 MB) at F1 0.7143, indicating that model size alone does not determine deepfake detection capability.
- Entropy-based thresholding outperformed confidence-based thresholding for M3, M8, and M9. This suggests that prediction uncertainty — rather than peak confidence — is a more

informative signal for detecting audio anomalies associated with synthetic generation in certain architectures.

5.3 Best Performing Model — M3 (AST-AudioSet)

The Audio Spectrogram Transformer (M3) achieved the highest performance across all metrics: accuracy 87%, precision 0.8364, recall 0.92, specificity 0.82, and F1 0.8762. Its entropy-based thresholding strategy at the inverted 55th percentile threshold (threshold = 0.80) proved highly effective. Unlike waveform-based models that process raw time-domain signals, AST converts audio to a 2D time-frequency spectrogram and applies patch-based ViT attention. Spectrogram patterns of synthetic audio — including overly smooth formant transitions, lack of natural prosodic variation, and frequency artifacts from vocoder processing — are well captured by this representation, explaining M3's superior performance.

5.4 Worst Performing Model — M2 (wav2vec2-superb-er)

M2 (superb/wav2vec2-base-superb-er) achieved the lowest accuracy (51%) and specificity (0.02), essentially predicting nearly all samples as fake. Its emotion recognition features — trained to distinguish categories like happy, angry, and neutral — provide no meaningful separation between real and synthetic audio at the acoustic level relevant to deepfake detection. This illustrates a fundamental limitation: task-irrelevant pre-training objectives do not always yield transferable features for binary deepfake detection.

5.5 Effect of Threshold Tuning

Prior to adaptive threshold tuning, preliminary experiments revealed that many models defaulted to single-class predictions, particularly all-fake, due to the score distribution characteristics of their output spaces. The percentile-based and fixed threshold search procedures successfully recovered meaningful binary predictions for all ten models. For M3, the optimal threshold of 0.80 was identified through a systematic sweep, showing that F1 peaked at this value and declined with both lower and higher thresholds. This result underscores that the choice of decision boundary is as important as model architecture when repurposing pre-trained models for new tasks.

5.6 Ensemble Analysis

The weighted ensemble achieved F1 of 0.6897 and accuracy of 55% — both below M3's individual performance. The ensemble's perfect recall (1.00) indicates that it never misses a fake sample, but its specificity of only 0.10 means it classifies 90% of real samples as fake, leading to an unacceptably high false alarm rate. This behavior reflects the ensemble's composition: eight of the ten component models have specificities below 0.45, collectively biasing the weighted vote toward the fake class. This finding demonstrates that ensemble methods are beneficial primarily when component models are individually strong and diverse, not when they are uniformly biased.

5.7 Limitations of Pre-trained Models

Several important limitations of the pre-trained approach emerged from this study. First, label space mismatch is the most significant challenge — none of the evaluated models were designed for

binary real/fake classification, requiring an indirect score-based inference strategy that inherently loses information compared to task-specific fine-tuning. Second, domain gap is evident: models trained on speech emotion, keyword spotting, or audio event detection encode acoustic features that may not align with the spectral and prosodic artifacts characteristic of deepfake audio. Third, the small evaluation set (100 samples) introduces statistical uncertainty in the reported metrics, particularly for models near the chance threshold. Fourth, the dataset limitation is notable — the garystafford dataset, while useful for benchmarking, may not represent the full diversity of real-world deepfake generation methods.

6. Conclusion

This study demonstrated that pre-trained transformer-based audio models can be repurposed for deepfake audio detection through adaptive score-based thresholding, without any task-specific fine-tuning. The Audio Spectrogram Transformer (AST) fine-tuned on AudioSet emerged as the strongest performer, achieving 87% accuracy and an F1 score of 0.8762 on a balanced 100-sample benchmark drawn from the garystafford/deepfake-audio-detection dataset. This result suggests that spectrogram-based representations capture audio anomalies indicative of synthetic speech more effectively than waveform-based self-supervised models in a zero-shot transfer setting.

The adaptive threshold tuning methodology — combining percentile-based, fixed, and median-based strategies with both confidence and entropy signals — proved essential for recovering meaningful binary predictions from models whose output label spaces are incompatible with the target task. Without this procedure, the majority of models would produce degenerate single-class predictions.

Contrary to typical expectations, the weighted ensemble did not outperform the best individual model. While the ensemble achieved perfect recall, its poor specificity (0.10) makes it unsuitable for real-world deployment without significant refinement. This finding highlights that ensemble benefits are contingent on the quality and diversity of component models — conditions not fully met in the zero-shot pre-trained setting.

The successful Gradio deployment of the best model (M3/AST) at a tuned threshold of 0.80 demonstrates the practical feasibility of building real-time deepfake audio detection systems using the HuggingFace ecosystem with minimal infrastructure.

7. Future Work

Several directions for extending this work are identified:

- Fine-tuning on deepfake-specific data: The most impactful improvement would be to fine-tune one or more of the evaluated models (particularly M3/AST or M6/WavLM-base) on a large deepfake audio dataset such as ASVspoof 2019 or FakeAVCeleb. This would eliminate the label space mismatch and likely push performance into the 95%+ accuracy range.

- Larger and more diverse evaluation sets: Evaluating on the full garystafford dataset (1,866 samples) and supplementing with additional datasets including different synthesis methods (GAN-based, neural vocoder, voice conversion) would improve the statistical robustness of conclusions.
- Multimodal deepfake detection: Combining audio deepfake detection with visual deepfake detection (using face-forgery detection models) into a unified audiovisual pipeline could significantly improve robustness against sophisticated multimodal deepfakes that manipulate both speech and video simultaneously.
- Exploration of newer models: More recent architectures such as Whisper (OpenAI), EnCodec (Meta), or WavLM-Large could be included in the comparison, as these models encode richer representations that may transfer more effectively to deepfake detection.
- Online learning and adaptation: Developing an online threshold adaptation mechanism that updates the decision boundary as new samples arrive could improve real-time detection performance in streaming environments.
- Explainability: Incorporating attention visualization and saliency maps to identify which acoustic features drive real/fake predictions would improve interpretability and support forensic analysis.
- Fine-grained ensemble design: Training a meta-learner (stacking) on held-out data to optimally combine model predictions — rather than simple F1-weighted voting — could better exploit complementary model behaviors.

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